1 Solution — GIAM 1.2 Problem 8

1.1 Problem

Use the second definition of "prime" to show that 35 is not a prime.

1.2 The Second Definition

From Section 1.2:

> *A prime is a quantity p such that whenever p is a factor of some product ab, then either p is a factor of a or p is a factor of b.*

That is, p is prime iff $p \mid ab \Rightarrow p \mid a \text{ or } p \mid b$. To show 35 is *not* prime, we exhibit one product ab that 35 divides while dividing neither a nor b.

1.3 The Counterexample

Since $35 = 5 \cdot 7$, take

$$a = 5, \qquad b = 7, \qquad ab = 35.$$

Then:

- $35 \mid 35$ — yes, 35 is a factor of the product $ab = 35$.
- $35 \nmid 5$ — 35 is *not* a factor of $a = 5$.
- $35 \nmid 7$ — 35 is *not* a factor of $b = 7$.

So 35 divides the product ab but divides *neither* a nor b, violating the second definition. Hence 35 is **not prime**. ■

1.4 Why It Works

As with 6 in Problem 7, the trick is that $35 = 5 \cdot 7$ is *composite*: it is built from two distinct prime factors. We give the factor 5 to a and the factor 7 to b. The product ab then carries *both* primes (so $35 \mid ab$), but $a = 5$ is missing the 7 and $b = 7$ is missing the 5, so 35 divides neither.

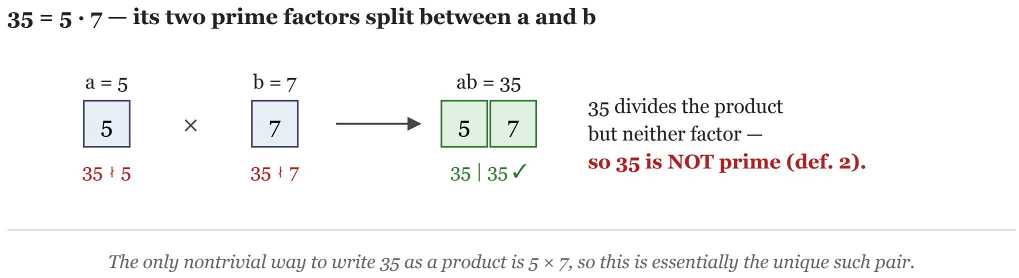

Figure 1.1: 35 deals its factor 5 to a and its factor 7 to b. The product ab = 35 contains both primes (35 | 35), but neither a = 5 nor b = 7 is divisible by 35. The only nontrivial factorization of 35 is 5 × 7, so this is essentially the unique counterexample of this form.

1.5 Remark — More Counterexamples

The product need not be exactly 35. Any a, b that split the prime factors 5 and 7 between them works:

a	b	ab	$35 \mid ab$?	$35 \mid a$?	$35 \mid b$?
5	7	35	yes	no	no
10	7	70	yes	no	no
5	14	70	yes	no	no
15	14	210	yes	no	no
25	21	525	yes	no	no

Table 1.1

Each row works because one number supplies a factor of 5 (but no 7) and the other supplies a factor of 7 (but no 5).

1.6 Remark — 35 Is a Semiprime, So the Pair Is Essentially Unique

The positive divisors of 35 are $1, 5, 7, 35$, so the only ways to write 35 as a product of two positive integers are

$$35 = 1 \times 35 \qquad \text{and} \qquad 35 = 5 \times 7.$$

The trivial factorization 1×35 is no help (since $35 \mid 35$), so $a = 5$, $b = 7$ is the **essentially unique** counterexample whose product equals 35 itself. A number like 35 — a product of *two distinct primes* — is called a **semiprime**; $6 = 2 \cdot 3$ from Problem 7 is another. Semiprimes are the "simplest" composites: they split into exactly two prime atoms, one for each of a and b.

1.7 Remark — Same Mechanism, Stated Once More in General

This is the identical phenomenon as Problem 7, and the general fact bears repeating: **every composite n fails the second definition.** If $n = de$ with $1 < d, e < n$, then $a = d$, $b = e$ gives

$$n \mid ab = n \quad \text{(trivially)}, \qquad \text{but } n \nmid d \text{ and } n \nmid e,$$

because d, e are positive and strictly smaller than n. For 35 this is the factorization $35 = 5 \cdot 7$.

Conversely, a genuine prime p *does* satisfy the definition — that is **Euclid's lemma** — so passing the second definition (together with being > 1) is exactly equivalent to being prime. The deeper significance of having a *second* definition at all — that "prime" and "irreducible" diverge in more general number systems such as $\mathbb{Z}[\sqrt{-5}]$ — is discussed at length in the remarks to Problem 7.